

Q.1.1 Let $|\psi_1\rangle = |0\rangle$, $|\psi_2\rangle = |+\rangle$

$$|1\rangle = \sqrt{2}|\psi_2\rangle - |\psi_1\rangle = -|\psi_1\rangle + \sqrt{2}|\psi_2\rangle$$

$\therefore$ Co-ordinates of $|1\rangle$ are $(-1, \sqrt{2})$

$$|-\rangle = \sqrt{2}|\psi_1\rangle - |\psi_2\rangle$$

$\therefore$ Co-ordinates of $|-\rangle$ are $(\sqrt{2}, -1)$



Q.1.2. $|\psi_1\rangle = u_{11}|0\rangle + u_{12}|+\rangle$

$$|\psi_2\rangle = u_{21}|0\rangle + u_{22}|+\rangle$$

$$\langle\psi_1|\psi_2\rangle = (\langle 0|u_{11}^* + \langle +|u_{12}^*)(u_{21}|0\rangle + u_{22}|+\rangle)$$

$$= u_{11}^* u_{21} \langle 0|0\rangle + u_{11}^* u_{22} \langle 0|+\rangle + u_{12}^* u_{21} \langle +|0\rangle + u_{12}^* u_{22} \langle +|+\rangle$$

$$= (u_{11}^* u_{21} + u_{12}^* u_{22}) + \frac{u_{11}^* u_{22} + u_{12}^* u_{21}}{\sqrt{2}}$$

$$= \begin{bmatrix} u_{11}^* & u_{12}^* \end{bmatrix} \begin{bmatrix} u_{21} & u_{22} \\ u_{22} & u_{21} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11}^* & u_{12}^* \end{bmatrix} \begin{bmatrix} 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} u_{21} \\ u_{22} \end{bmatrix}$$

$$\therefore \langle v_1 | v_2 \rangle = \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix}^\dagger \begin{bmatrix} 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix}$$

$$\langle -1 | 1 \rangle = \begin{bmatrix} \sqrt{2} & -1 \end{bmatrix} \begin{bmatrix} 1 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1 \end{bmatrix} \begin{bmatrix} -1 \\ \sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1/\sqrt{2} \end{bmatrix} = \underline{\underline{-1/\sqrt{2}}}$$

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Q.2. Note that $\frac{|0\rangle}{\sqrt{2}} + \frac{|1\rangle}{\sqrt{2}} - |+\rangle = 0$

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Hence any vector $|v\rangle = \langle 0|v\rangle |0\rangle + \langle 1|v\rangle |1\rangle$ can also be expressed as

$$|v\rangle = \langle 0|v\rangle |0\rangle + \langle 1|v\rangle |1\rangle + x \left[\frac{|0\rangle}{\sqrt{2}} + \frac{|1\rangle}{\sqrt{2}} - |+\rangle \right]$$

for any $x \in \mathbb{C}$

$$\therefore |v\rangle = \left(\langle 0|v\rangle + \frac{x}{\sqrt{2}} \right) |0\rangle + \left(\langle 1|v\rangle + \frac{x}{\sqrt{2}} \right) |1\rangle - x |+\rangle$$

is a valid representation for $\forall x \in \mathbb{C}$

Thus, there are uncountably (infinite) many possible representations in the given set for any vector $|0\rangle$

$$|- \rangle = \frac{|0\rangle}{\sqrt{2}} + \frac{|1\rangle}{\sqrt{2}} - 0 \cdot |+\rangle, (\alpha = 0)$$

$$\equiv -\sqrt{2} |1\rangle + |+\rangle, (\alpha = -1)$$

Similarly $|1\rangle = 0 \cdot |0\rangle + |1\rangle + 0 \cdot |+\rangle, (\alpha = 0)$

$$\textcircled{2} \equiv -|0\rangle + \sqrt{2} |+\rangle, (\alpha = -\sqrt{2})$$

Q.3. $\sigma_x = |0\rangle\langle 1| + |1\rangle\langle 0|$

$$|0\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}}, |1\rangle = \frac{|+\rangle - |-\rangle}{\sqrt{2}} \quad \textcircled{4}$$

$$\sigma_x = \left(\frac{|+\rangle + |-\rangle}{\sqrt{2}} \right) \left(\frac{\langle +| - \langle -|}{\sqrt{2}} \right) + \left(\frac{|+\rangle - |-\rangle}{\sqrt{2}} \right) \left(\frac{\langle +| + \langle -|}{\sqrt{2}} \right)$$

$$= \frac{1}{2} [2|+\rangle\langle +| - 2|-\rangle\langle -|]$$

$$= |+\rangle\langle +| - |-\rangle\langle -|$$

Q.4. 1) $\sigma_x |+\rangle = +1 |+\rangle$ Eigenvectors
 $\sigma_x |-\rangle = -1 |-\rangle$ Eigenvalues
 $\Rightarrow \sigma_x = |+\rangle\langle+| - |-\rangle\langle-|$

(3) each except Q-4 Q-4

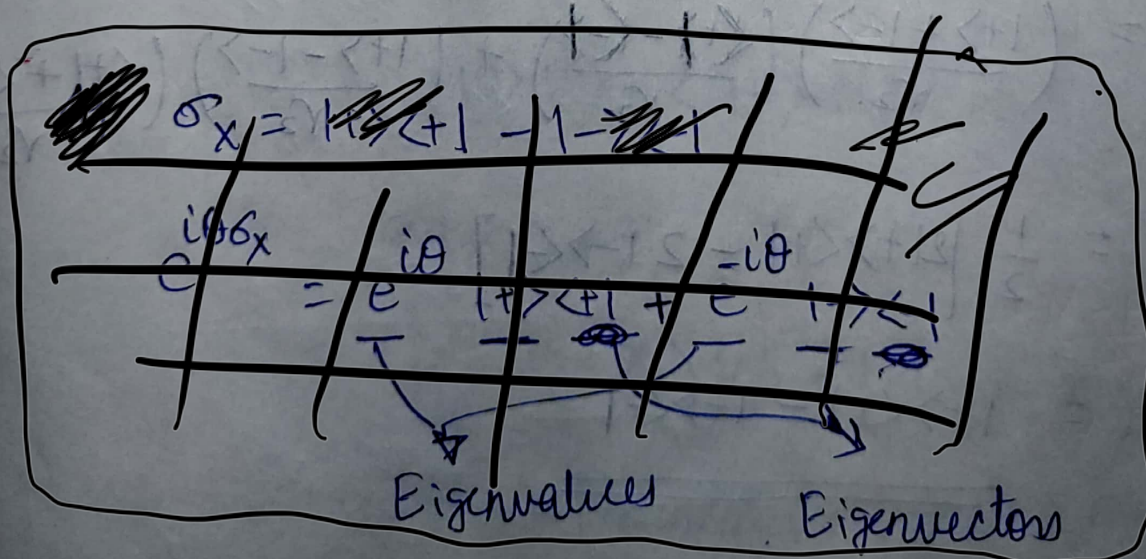
2) $\sigma_y \begin{bmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{bmatrix} = +1 \cdot \begin{bmatrix} 1/\sqrt{2} \\ i/\sqrt{2} \end{bmatrix} \rightarrow |+i\rangle, |-i\rangle$ are eigenvectors
 $\sigma_y \begin{bmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{bmatrix} = -1 \cdot \begin{bmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{bmatrix}$ with eigenvalues ± 1 respectively.

$\sigma_y = |+i\rangle\langle+i| - |-i\rangle\langle-i|$

3) $\sigma_z |0\rangle = +1 \cdot |0\rangle$

$\sigma_z |1\rangle = -1 \cdot |1\rangle$

$\Rightarrow \sigma_z = |0\rangle\langle 0| - |1\rangle\langle 1|$



4) Note that $\vec{n} \cdot \vec{\sigma} = \begin{bmatrix} \cos\theta & \sin\theta e^{-i\phi} \\ \sin\theta e^{i\phi} & -\cos\theta \end{bmatrix}$

$$(\vec{n} \cdot \vec{\sigma})^\dagger = \vec{n} \cdot \vec{\sigma}$$

$$\text{and } (\vec{n} \cdot \vec{\sigma})^\dagger (\vec{n} \cdot \vec{\sigma}) = I$$

$\Rightarrow \vec{n} \cdot \vec{\sigma}$ is unitary matrix, ~~Hermitian~~

$$\text{Furthermore as } (\vec{n} \cdot \vec{\sigma})^2 = I \text{ \& } \vec{n} \cdot \vec{\sigma} \neq \pm I$$

$\Rightarrow$ the eigenvalues of $\vec{n} \cdot \vec{\sigma}$ are ± 1 & -1 .

$$\left[\begin{array}{l} \vec{n} \cdot \vec{\sigma} = e^{i\theta_+} |\psi_+\rangle \langle \psi_+| + e^{i\theta_-} |\psi_-\rangle \langle \psi_-| \text{ as } \vec{n} \cdot \vec{\sigma} \text{ is diagonalizable} \\ (\vec{n} \cdot \vec{\sigma})^\dagger = e^{-i\theta_+} |\psi_+\rangle \langle \psi_+| + e^{-i\theta_-} |\psi_-\rangle \langle \psi_-| \\ \text{as } (\vec{n} \cdot \vec{\sigma})^\dagger = \vec{n} \cdot \vec{\sigma} \Rightarrow e^{i\theta_+} = e^{-i\theta_+} \text{ \& } e^{i\theta_-} = e^{-i\theta_-} \\ \Rightarrow e^{2i\theta_+} = 1 \text{ \& } e^{2i\theta_-} = 1 \\ \Rightarrow \theta_+, \theta_- = 0 \text{ or } \pi \end{array} \right]$$

Both eigenvalues can't be equal as that would imply $\vec{n} \cdot \vec{\sigma} = \pm I$, hence $\theta_+ = 0, \theta_- = \pi$

5) $\therefore \vec{n} \cdot \vec{\sigma} = |\psi_+\rangle \langle \psi_+| - |\psi_-\rangle \langle \psi_-|$

Consider $|\psi_+\rangle = \begin{bmatrix} \cos \alpha/2 \\ e^{i\beta} \sin \alpha/2 \end{bmatrix}$ for some α, β
in Bloch ^{vector} representation

$$\vec{n} \cdot \vec{\sigma} |\psi_+\rangle = +1 \cdot |\psi_+\rangle$$

$$\Rightarrow \begin{bmatrix} \cos \theta \cos \alpha/2 + e^{i(\beta-\phi)} \sin \theta \sin \alpha/2 \\ e^{i\phi} (\sin \theta \cos \alpha/2 - \cos \theta \sin \alpha/2) \end{bmatrix} = \begin{bmatrix} \cos \alpha/2 \\ e^{i\beta} \sin \alpha/2 \end{bmatrix}$$

$$\Rightarrow \cos \theta \cos \alpha/2 + e^{i(\beta-\phi)} \sin \theta \sin \alpha/2 = \cos \alpha/2$$

$$\text{and } \sin \theta \cos \alpha/2 e^{i(\phi-\beta)} - \cos \theta \sin \alpha/2 = \sin \alpha/2$$

$$\Rightarrow \beta = \phi \text{ and } \alpha = \theta$$

$$\therefore |\psi_+\rangle = \begin{bmatrix} \cos \theta/2 \\ e^{i\phi} \sin \theta/2 \end{bmatrix}$$

Similarly

$$|\psi_-\rangle = \begin{bmatrix} \cos(\frac{\pi-\theta}{2}) \\ e^{i(\phi+\pi)} \sin(\frac{\pi-\theta}{2}) \end{bmatrix} = \begin{bmatrix} \sin \theta/2 \\ -e^{i\phi} \cos \theta/2 \end{bmatrix}$$

5 ~~$(\sigma_x \otimes \sigma_x)(|1\rangle \otimes |1\rangle)$~~

As $\sigma_x | \pm \rangle = \pm 1 \cdot | \pm \rangle$

$\therefore$ Eigenvectors of $\sigma_x \otimes \sigma_x$ are $|+\rangle \otimes |+\rangle$, $|+\rangle \otimes |-\rangle$, $|-\rangle \otimes |+\rangle$ and $|-\rangle \otimes |-\rangle$. The corresponding eigenvalues are $+1, -1, -1$ and $+1$.

$$\left. \begin{aligned} (\sigma_x \otimes \sigma_x)(|+\rangle \otimes |-\rangle) &= (\sigma_x |+\rangle \otimes \sigma_x |-\rangle) \\ &= (+1 \cdot |+\rangle) \otimes (-1 \cdot |-\rangle) = -|+\rangle \otimes |-\rangle \end{aligned} \right\}$$

6 As $\sigma_y | \pm i \rangle = \pm 1 \cdot | \pm i \rangle$ + $\sigma_x | \pm \rangle = \pm 1 \cdot | \pm \rangle$

$\therefore$ Eigenvectors of $\sigma_x \otimes \sigma_y$ are $|+\rangle \otimes |i\rangle$, $|+\rangle \otimes |-i\rangle$, $|-\rangle \otimes |i\rangle$ + $|-\rangle \otimes |-i\rangle$. The corresponding eigenvalues are $+1, -1, -1$ + $+1$.

7 Note that $|u_1\rangle \langle u_2 | u_2^\perp \rangle = 0 \cdot |u_2^\perp\rangle$, where $|u_2^\perp\rangle$ is unit vector ~~perpend~~ orthogonal to $|u_2\rangle$.
 $\therefore$ $|u_2^\perp\rangle$ is an eigenvector with eigenvalue 0.

Also $|u_1\rangle \langle u_2 | u_1 \rangle = \langle u_2 | u_1 \rangle |u_1\rangle \Rightarrow$ $|u_1\rangle$ is an eigenvector with eigenvalue $\langle u_2 | u_1 \rangle$.

Q.5.1. $AA^\dagger = BB^\dagger = I$

$$(A+B)^\dagger (A+B) = AA^\dagger + BA^\dagger + AB^\dagger + BB^\dagger \\ = 2I + BA^\dagger + AB^\dagger \neq I$$

$\therefore (A+B)$ need not be unitary

$$(AB)(AB)^\dagger = AB(B^\dagger A^\dagger) = ABB^\dagger A^\dagger \\ = AIA^\dagger = AA^\dagger = I$$

$\therefore AB$ is unitary.

$$(A \otimes B)(A \otimes B)^\dagger = (A \otimes B)(A^\dagger \otimes B^\dagger) \\ = AA^\dagger \otimes BB^\dagger = I \otimes I$$

$\therefore A \otimes B$ is unitary.

2. $A^\dagger = A, B^\dagger = B$

$$(A+B)^\dagger = A^\dagger + B^\dagger = (A+B)$$

$\therefore A+B$ is Hermitian

$(AB)^\dagger = B^\dagger A^\dagger = BA$ which need not be equal to AB
 $\therefore AB$ is not necessarily Hermitian.

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger = A \otimes B$$

$\therefore (A \otimes B)$ is Hermitian.

3. $A^\dagger A = A A^\dagger, B^\dagger B = B B^\dagger$

$$\begin{aligned} (A+B)^\dagger (A+B) &= (A^\dagger + B^\dagger)(A+B) = A^\dagger A + A^\dagger B + B^\dagger A + B^\dagger B \\ &= (A A^\dagger + B B^\dagger) + (A^\dagger B + B^\dagger A) \\ &= (A A^\dagger + B B^\dagger + A B^\dagger + B A^\dagger) + (A^\dagger B + B^\dagger A - A B^\dagger - B A^\dagger) \\ &= (A+B)(A+B)^\dagger + (A^\dagger B - B A^\dagger + B^\dagger A - A B^\dagger) \end{aligned}$$

$\therefore (A+B)^\dagger (A+B) \neq (A+B)(A+B)^\dagger$ in general.

Hence $(A+B)$ need not be normal.

$$(AB)^\dagger AB = B^\dagger A^\dagger AB$$

$$(AB)(AB)^\dagger = AB B^\dagger A^\dagger$$

$\therefore (AB)^\dagger AB \neq AB(AB)^\dagger$ in general

Hence AB need not be normal

$$\begin{aligned} (A \otimes B)^\dagger (A \otimes B) &= (A^\dagger \otimes B^\dagger)(A \otimes B) = (A^\dagger A) \otimes (B^\dagger B) = A A^\dagger \otimes B B^\dagger \\ &= (A \otimes B)(A \otimes B)^\dagger \end{aligned}$$

$\therefore A \otimes B$ is normal.